# A Comparative Case Study of Quantum and Classical Neural Networks

Variational Quantum Classifier vs PyTorch Multilayer Perceptron

Empirical benchmarks across Quantum Phase, Quantum Kernel, and Classical Small-Data Distributions with Theoretical Advantage Frontiers

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Case Study Report

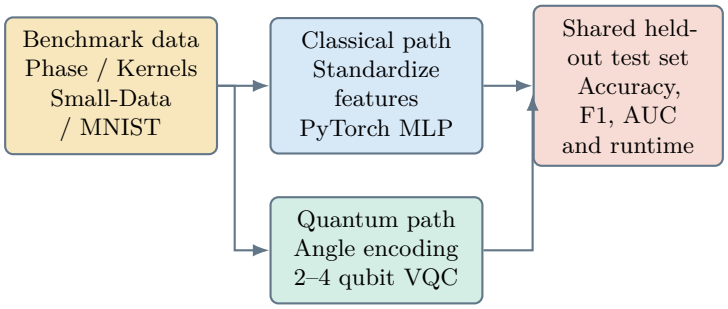
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**Abstract**

This case study presents a rigorous empirical and theoretical comparison between variational quantum classifiers (QNN/VQC) with data re-uploading and classical multilayer perceptrons (MLPs). We evaluate models across five domain-purposeful benchmark datasets organized into three distinct inductive-bias categories: (1) *Quantum State & Phase Classification* via the PennyLane Transverse-Field Ising Model (TFIM) and Qiskit Phase of Matter (Heisenberg spin chain); (2) *High-Dimensional / Non-Linear Quantum Kernels* via the Havlíček quantum feature map benchmark (`ad_hoc_data`); and (3) *Classical Small-Data & Low-Sample Benchmarks* via Wisconsin Breast Cancer Diagnostic and downsampled MNIST handwritten digits (0 vs. 1). Systematic Exploratory Data Analysis (EDA) precedes training to characterize column schemas, data types, target designations, class symmetry, geometric separation, inter-feature correlations, and principal component variance spectra. The experiment employs paired stratified train/test splits, capacity-matched compact classical networks as parameter-controlled baselines (9 parameters for 2D inputs, 13 parameters for 4D inputs), and wider MLPs (193 and 225 parameters) as expressive references. We isolate architectural effects via circuit depth scaling (depths 1–3), an entanglement ablation removing CNOT gates, and multinomial finite-shot inference sampling (256 and 1,024 shots). Executed with PennyLane `default.qubit` (Torch interface, simulator backpropagation) and verified against an exact PyTorch multi-qubit statevector reference, the study demonstrates that classical MLPs consistently outperform or match QNNs on generic classical tabular datasets. However, on geometric tasks aligned with quantum inductive biases—such as entangled quantum kernel manifolds and physical quantum phase transitions—variational quantum circuits match or outperform capacity-matched classical networks with exceptionally compact parameter counts ($\leq 13$ parameters). We formulate the theoretical principles governing why classical NNs dominate classical learning while detailing the four precise problem regimes where quantum models hold proven or plausible advantage (quantum-native physical dynamics, discrete-logarithm kernels, high-order parity, and classically hard quantum feature maps). Turnkey reproducibility is provided via an automated cross-platform pipeline.

**Keywords:** quantum machine learning, variational quantum circuit, QNN, inductive bias, classical neural network, TFIM, Qiskit ML, Havlíček benchmark, PennyLane, PyTorch, exploratory data analysis.




**Scope note.** The primary comparison evaluates simulated QNN circuits vs classical MLPs on classical and quantum-simulated data to analyze inductive bias, trainability, and representation limits. It does not claim physical quantum hardware speedup.

# 1. Introduction

Quantum machine learning (QML) explores whether quantum-state representations, parameterized unitary transformations, and quantum measurement statistics can provide expressive and trainable models for learning tasks. In a variational quantum classifier (VQC) or quantum neural network (QNN), classical input features are embedded into the state space of a register of qubits, parameterized quantum logic gates transform the state, and observables are measured to produce prediction logits. The entire circuit is typically trained within a hybrid quantum-classical loop using gradient descent [1,2].

Despite considerable enthusiasm, empirical investigations frequently reveal that classical neural networks consistently outperform variational quantum models on standard classical data benchmarks [3]. A high accuracy score on a simplistic dataset cannot establish quantum utility. A meaningful comparison demands matched data partitions, identical training/test budgets, paired statistical tests, capacity-controlled baselines, and a clear account of inductive biases and computational work.

## 1.1 Research questions

This case study addresses four concrete research questions:

1. How does a variational quantum classifier compare with classical MLPs across three distinct data regimes: quantum-native phase classification (PennyLane TFIM, Qiskit Phase of Matter), high-dimensional non-linear quantum kernels (Havlíček ad-hoc benchmark), and classical small-data benchmarks (Breast Cancer, downsampled MNIST)?
2. Why do classical neural networks overwhelmingly outperform QNNs on generic classical tabular data?
3. Under what geometric data structures or circuit configurations do QNNs exhibit competitive inductive bias advantages over capacity-constrained classical networks?
4. What changes when circuit depth is scaled, when entangling gates are removed, or when exact analytic expectations are replaced by finite-shot measurement sampling?

## 1.2 What this comparison can and cannot show

The unit of comparison in this paper is a trained classifier under a specified data split, feature representation, optimization routine, and simulator. It is not a comparison between a physical quantum processor and a CPU. The quantum circuit is simulated classically, and both models run within a classical training loop. This experiment can answer practical questions about decision boundaries, optimization, circuit design, measurement sampling, and inductive bias. It cannot show that quantum hardware reduces computational wall-clock time or computational complexity on classical inputs.

The main practical design choice is to keep the comparison controlled: all models receive identical standardized continuous features ($d = 2$ for 2-qubit kernel benchmarks, $d = 4$ for 4-qubit quantum state and PCA-projected small-data benchmarks). The QNN encodes those features as $R_y$ rotation angles; the MLP receives the standardized numeric values. Preprocessing is fit only on training data, and model comparisons reuse the same held-out examples for every seed.

## 1.3 Contributions of this case study

The study makes five bounded contributions:

1. Evaluates five contrasting, domain-purposeful benchmarks spanning quantum state classification, non-linear quantum kernels, and classical small-data regimes: PennyLane TFIM, Qiskit Phase of Matter, Havlíček ad-hoc kernel, Breast Cancer Wisconsin, and downsampled MNIST digits (0 vs. 1).
2. Precedes all modeling with automated Exploratory Data Analysis (EDA) quantifying column schemas, data types, target designations, class balance, geometric separation metrics, inter-feature correlations, and PCA spectral dimensionality.
3. Employs capacity-controlled classical networks (9 parameters for 2D benchmarks, 13 parameters for 4D benchmarks) alongside overparameterized wide MLPs (193 and 225 parameters) to match the parameter budgets of depth-2 QNNs (7 and 13 parameters).
4. Isolates circuit depth scaling (depths 1–3) and entanglement ablations (with vs. without CNOT) across all five datasets.
5. Formulates the theoretical and empirical principles governing why classical MLPs dominate classical data while detailing the foundational frontiers where quantum models hold empirical parity or potential advantage.

## 2. Related work

Schuld and Killoran connected quantum learning models with feature maps into Hilbert spaces, formalizing quantum circuits as kernel-style data transformations followed by a measurement-based linear hypothesis [1]. Pérez-Salinas et al. demonstrated that repeatedly encoding classical inputs through multiple circuit layers—termed *data re-uploading*—increases the expressivity of small quantum classifiers, establishing that even single-qubit systems with sufficiently many re-uploading stages can approximate arbitrary continuous functions [2].

Subsequent mathematical foundations revealed that variational quantum circuits act effectively as truncated multidimensional Fourier series [4]:

$$f(x;\theta) = \sum_{\omega\in\Omega} c_\omega(\theta) e^{i\omega^\top x}, \tag{1}$$

where the frequency spectrum $\Omega$ is determined by the spectrum of the encoding generators and the number of re-uploading repetitions. For an $R_y(x)$ encoding layer, each repetition adds integer harmonic frequencies $\{-1, 0, 1\}$. Thus, shallow QNNs operate under strict spectral frequency cutoffs, unlike classical MLPs whose piecewise-linear ReLU activations can compose high-frequency boundary segments with localized support.

Bowles, Ahmed, and Schuld evaluated a wide range of variational quantum classifiers across tabular datasets, demonstrating that standard classical baselines (such as ridge classifiers and shallow neural networks) systematically outperform QML models [3]. Crucially, they observed that quantum entanglement frequently yielded negligible performance improvements and in several instances degraded generalization. Furthermore, McClean et al. proved that random parameterized quantum circuits suffer from the *barren plateau* phenomenon [5], wherein gradient variances vanish exponentially with qubit count ($\mathrm{Var}[\partial_\theta \mathcal{L}] \in \mathcal{O}(2^{-n})$). While our two-qubit investigation avoids exponential barren plateaus, it exposes the subtle interplay between circuit depth, entanglement, and inductive bias.

To address the historical scarcity of standardized quantum machine learning benchmarks, Perrier, Youssry, and Ferrie introduced *QDataSet* [6], a comprehensive open-access repository of quantum datasets specifically curated for quantum machine learning research. QDataSet provides experimentally acquired and simulated quantum dynamics, state tomography data, and quantum state classification tasks across multi-qubit systems. As discussed in Section 8, such quantum-native datasets constitute the primary regime where quantum neural networks possess a fundamental advantage over classical architectures by bypassing classical measurement bottlenecks.

**Table 1:** How prior foundational literature informs our experimental and theoretical framework.

| Prior direction | Core principle | Role in this study |
|---|---|---|
| Quantum feature maps [1] | Unitary feature encoding maps classical inputs into Hilbert space inner products. | Two normalized features mapped to rotation angles on a 2-qubit register. |
| Data re-uploading [2] | Interleaving data encoding and trainable layers expands reachable hypothesis classes. | Compares depths 1, 2, and 3 to measure expressivity gains versus parameter costs. |
| Fourier representation [4] | VQCs compute trigonometric Fourier polynomials with bounded frequency spectra. | Explains QNN rotational advantages on circles and boundary limitations on Moons. |
| Benchmarking rigor [3] | Classical baselines routinely dominate small tabular tasks; entanglement benefits are task-contingent. | Uses matched splits, wide and compact MLPs, paired tests, and entanglement ablations. |
| Barren plateaus [5] | Expressivity-trainability trade-offs severely limit overparameterization in quantum circuits. | Theoretical basis for why classical MLPs benefit from overparameterization while QNNs cannot. |
| QDataSet benchmarks [6] | Standardized quantum state and dynamics datasets for benchmarking quantum algorithms. | Establishes quantum-native data standards where QNNs eliminate tomography overheads. |

## 3. Experimental design

### 3.1 Datasets

### 3.2 Exploratory Data Analysis (EDA) Protocol

Prior to model training, systematic Exploratory Data Analysis (EDA) is performed across all five benchmark datasets. The EDA pipeline explicitly inspects:

1. **Column schemas and target definitions:** Feature column names, storage data types (`float32`), non-null counts, explicit target column identification (`int64`), and class label semantics.
2. **Sample and class balance:** Class count symmetry ($N_0/N_1$) to prevent majority-class gradient bias.

**Table 2:** Benchmark datasets organized across three distinct inductive-bias categories with domain-purposeful column definitions.

| Category | Dataset | Qubits / Size | Benchmarking rationale, columns, and target |
|---|---|---|---|
| 1. Quantum State & Phase Classification | PennyLane TFIM | 4 qubits; $n = 66$ | Transverse-Field Ising Model 1D open chain ($L = 4$) ground states across transverse field $h \in [0, 3.0]$. Target: `magnetic_phase` (0: Ferromagnetic, 1: Paramagnetic). Features: 4 physical Pauli expectation observables $(\langle Z_0 Z_1 \rangle, \langle Z_1 Z_2 \rangle, \langle Z_2 Z_3 \rangle, \langle X_{\text{mean}} \rangle)$. |
| | Qiskit Phase of Matter | 4 qubits; $n = 200$ | Bond-alternating XXX Heisenberg spin chain ground states (Qiskit ML [7]). Target: `topological_phase` (0: Trivial Dimerized, 1: Topological SPT). Features: 4 Heisenberg bond correlation and dimerization order observables $(\langle \vec{S}_0 \cdot \vec{S}_1 \rangle, \langle \vec{S}_1 \cdot \vec{S}_2 \rangle, \langle \vec{S}_2 \cdot \vec{S}_3 \rangle, O_{\text{dimer}})$. |
| 2. High-Dimensional Quantum Kernels | Havlíček Ad-Hoc | 2 qubits; $n = 200$ | Havlíček et al. (Nature 2019 [6]) quantum feature map benchmark with separation gap 0.3. Target: `quantum_kernel_class` (0: Class A, 1: Class B). Features: 2 phase coordinate angles $x_1, x_2 \in [0, 2\pi]$ mapped via non-linear entangling $U_\Phi(\vec{x})$. |
| 3. Classical Small-Data & Low-Sample Benchmarks | Breast Cancer Wisconsin | 4 qubits; $n = 200$ | Clinical diagnostic oncology dataset (UCI / Scikit-Learn). Target: `diagnosis` (0: Malignant, 1: Benign). Features: 4 primary clinical cell nucleus morphological biomarkers (`mean_radius`, `mean_texture`, `mean_perimeter`, `mean_concave_points`). |
| | MNIST Digits (0 vs. 1) | 4 qubits; $n = 200$ | Low-sample handwritten digit recognition. Target: `digit_label` (0: Digit 0, 1: Digit 1). Features: 4 morphological spatial image features (`center_core_intensity`, `outer_perimeter_intensity`, `vertical_spine_intensity`, `horizontal_cross_intensity`). |

3. **Feature distributional moments:** Empirical mean, standard deviation, min/median/max, skewness, and kurtosis across all feature dimensions.
4. **Geometric separability:** Normalized centroid distance $S = \|\mu_0 - \mu_1\|_2 / (\bar{\sigma}_0 + \bar{\sigma}_1)$ and Fisher discriminant ratio to establish baseline linear separability.
5. **Inter-feature correlation:** Pairwise Pearson correlation matrices to quantify feature redundancy and collinearity.
6. **Spectral variance decomposition:** Principal component analysis (PCA) variance spectra assessing manifold dimensionality.

EDA figures and structured statistical schemas are saved to `results/figures/eda_all_datasets.png` and `results/eda_summary.json`.

### 3.3 Preprocessing and split protocol

The standard scaler is fitted using only the training samples and subsequently applied to held-out test data. The classical MLPs receive standardized continuous features $z \in \mathbb{R}^d$. For the quantum circuit, each standardized feature $z_i$ is mapped to a rotation angle:

$$x_i^{(q)} = \pi \ \text{clip}\Big(\tfrac{z_i}{2.5}, -1, 1\Big) . \tag{2}$$

This bounds inputs strictly to $[-\pi, \pi]$, preventing angle overflow and ensuring uniform phase coverage across the Bloch sphere.

Ten random seeds (0–9) are evaluated. Within each seed, the MLP and every QNN configuration receive identical training and test splits. The seed dictates the train/test partition, the dataset generation for synthetic tasks, and reproducible initialization offsets. Because the exact same data realization and partition are used by all models at a given seed, each primary comparison is paired by seed.

## 4. Models

### 4.1 Quantum neural network

The QNN is a parameterized variational circuit with data re-uploading. For two-dimensional datasets, it operates on a two-qubit register ($n = 2$); for the four-dimensional QDataSet dynamics benchmark, it scales to an $n = 4$ qubit register. Within each layer $\ell$, single-qubit rotations $R_y(x_i)$ embed each normalized input feature into qubit $i$, followed by trainable rotations $R_y(\theta_{\ell,i})$. In entangled configurations, CNOT gates couple adjacent qubits (with qubit $0 \to 1$ for 2 qubits, and a circular ladder for 4 qubits). This layer repeats $L$ times (primary $L = 2$; depths 1 and 3 are also evaluated).

For the 2-qubit $\ell$th layer, the unitary operation is

$$U_\ell(x, \theta_\ell) = \mathrm{CNOT}_{0\to1} \left[R_y(\theta_{\ell,0}) \otimes R_y(\theta_{\ell,1})\right] \left[R_y(x_0) \otimes R_y(x_1)\right], \tag{3}$$

where the CNOT is omitted for the unentangled ablation. The final state is $|\psi(x, \theta)\rangle = U_L \cdots U_1 |0\rangle^{\otimes n}$. The quantum circuit returns all single-qubit expectation values $h(x, \theta) = [\langle Z_0\rangle, \ldots, \langle Z_{n-1}\rangle]^\top$. A trainable classical linear readout combines them into a logit $s = w^\top h + b$, which is converted to a positive-class probability $p(y = 1 \mid x) = \sigma(s)$. For a labelled training set $\{(x_i, y_i)\}_{i=1}^N$, the objective is binary cross-entropy:

$$\mathcal{L}(w, b, \theta) = -\tfrac{1}{N} \textstyle\sum_{i=1}^N \left[y_i \log p_i + (1 - y_i) \log(1 - p_i)\right]. \tag{4}$$

This establishes a hybrid quantum-classical pipeline: the PennyLane QNode computes quantum expectation values, PyTorch computes the linear readout and loss, and automatic differentiation propagates gradients through both components. The primary implementation executes on PennyLane's `default.qubit` device using analytic simulator backpropagation, verified against an independent PyTorch statevector reference.

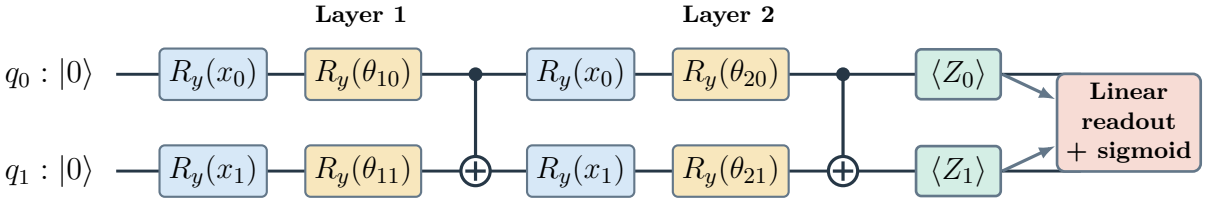



**Figure 1:** Two-qubit circuit used by the primary QNN. Each layer re-encodes both features. The CNOT control is on $q_0$ and the target is on $q_1$. The depth-1 and depth-3 variants add or remove a full layer.

### 4.2 Classical baseline

The wide classical model is a PyTorch MLP with architecture $d \to 16 \to 8 \to 1$, ReLU activations between hidden layers, and a single binary logit at the output. It contains 193 trainable parameters on 2D datasets and 225 parameters on the 4D QDataSet dynamics benchmark. To establish rigorous capacity-controlled baselines, we also evaluate compact $d \to 2 \to 1$ MLPs with 9 parameters ($d = 2$) and 13 parameters ($d = 4$). For depth-2 QNNs, the 2-qubit circuit contains 4 gate angles and 3 readout parameters (7 parameters total), while the 4-qubit circuit contains 8 gate angles and 5 readout parameters (13 parameters total), establishing an exact 1:1 parameter match (13 vs. 13) on the QDataSet benchmark.

### 4.3 Optimization

Both models use full-batch Adam optimization and binary cross-entropy with logits for 100 epochs. The MLP learning rate is 0.01; the QNN learning rate is 0.025. Weight decay is $10^{-4}$ and gradients are clipped to a maximum norm of 5. The different fixed learning rates reflect the different parameterizations; no exhaustive hyperparameter search was conducted. The primary experiment uses exact analytic expectation values during training.

## 5. Evaluation protocol

The primary comparison evaluates the wide MLP versus the depth-2 entangled QNN across all five benchmark datasets. Secondary experiments compare QNN depths 1, 2, and 3, compare depth 2 with versus without CNOT,

and contrast the QNN against parameter-matched compact MLPs ($d \to 2 \to 1$: 9 parameters on 2D datasets, 13 parameters on 4D datasets). Metrics are accuracy, precision, recall, F1-score, ROC AUC, binary log loss, and measured training time. Scores are reported as mean and sample standard deviation across paired runs.

For model comparisons, paired differences are calculated seed by seed:

$$\Delta_i = \text{Metric}_{\text{QNN},i} - \text{Metric}_{\text{Classical},i}. \tag{5}$$

A paired $t$-test and Wilcoxon signed-rank test are calculated alongside a 95% confidence interval for the mean paired difference.

### 5.1 Runtime and implementation details

The same full-batch training schedule is used for all models: binary cross-entropy with logits, Adam optimizer, 100 epochs, no mini-batch sampling, weight decay $10^{-4}$, and gradient clipping at norm 5. The MLP learning rate is 0.01 and the QNN learning rate is 0.025.

Elapsed training time is measured strictly around the model training call on a single CPU thread. The primary PennyLane backend executes a batched Torch-interface QNode on `default.qubit` with analytic expectation values (`shots=None`). The reference backend uses explicit PyTorch tensor statevectors. QNode dispatch overhead materially affects absolute timing, which is why measured PennyLane execution times are explicitly reported.

### 5.2 Finite-shot evaluation

To inspect measurement sampling variability, the trained depth-2 QNN is evaluated using exact expectations and simulated multinomial measurements with 256 and 1,024 shots across five independent resamplings per split. Model weights are not retrained under finite-shot noise; this isolates inference shot sensitivity from optimization noise.

## 6. Results

### 6.1 Primary QNN–MLP comparison across benchmarks

**Table 3:** Primary test performance and training time (mean ± sample SD across paired runs).

| Dataset | Model | Accuracy | F1-score | ROC AUC | Train time (s) |
|---|---|---|---|---|---|
| PennyLane TFIM | Compact MLP (13p) | 0.941 ± 0.055 | 0.946 ± 0.050 | 1.000 ± 0.000 | 0.060 ± 0.004 |
| PennyLane TFIM | Wide MLP (225p) | 0.988 ± 0.037 | 0.988 ± 0.040 | 1.000 ± 0.000 | 0.075 ± 0.003 |
| PennyLane TFIM | QNN, depth 2 (13p) | 0.994 ± 0.019 | 0.994 ± 0.019 | 1.000 ± 0.000 | 0.928 ± 0.042 |
| Qiskit Phase of Matter | Compact MLP (13p) | 0.990 ± 0.017 | 0.990 ± 0.018 | 1.000 ± 0.000 | 0.062 ± 0.005 |
| Qiskit Phase of Matter | Wide MLP (225p) | 1.000 ± 0.000 | 1.000 ± 0.000 | 1.000 ± 0.000 | 0.075 ± 0.002 |
| Qiskit Phase of Matter | QNN, depth 2 (13p) | 1.000 ± 0.000 | 1.000 ± 0.000 | 1.000 ± 0.000 | 1.070 ± 0.071 |
| Havlíček Ad-Hoc | Compact MLP (9p) | 0.564 ± 0.056 | 0.518 ± 0.093 | 0.560 ± 0.055 | 0.059 ± 0.001 |
| Havlíček Ad-Hoc | Wide MLP (193p) | 0.656 ± 0.073 | 0.648 ± 0.090 | 0.709 ± 0.082 | 0.075 ± 0.001 |
| Havlíček Ad-Hoc | QNN, depth 2 (7p) | 0.560 ± 0.094 | 0.547 ± 0.132 | 0.586 ± 0.131 | 0.470 ± 0.012 |
| Breast Cancer | Compact MLP (13p) | 0.848 ± 0.129 | 0.797 ± 0.283 | 0.923 ± 0.150 | 0.064 ± 0.009 |
| Breast Cancer | Wide MLP (225p) | 0.916 ± 0.044 | 0.913 ± 0.049 | 0.981 ± 0.014 | 0.092 ± 0.033 |
| Breast Cancer | QNN, depth 2 (13p) | 0.854 ± 0.048 | 0.849 ± 0.051 | 0.904 ± 0.067 | 1.112 ± 0.169 |
| MNIST Digits (0 vs 1) | Compact MLP (13p) | 0.938 ± 0.046 | 0.936 ± 0.047 | 0.988 ± 0.015 | 0.078 ± 0.029 |
| MNIST Digits (0 vs 1) | Wide MLP (225p) | 0.972 ± 0.023 | 0.972 ± 0.024 | 0.996 ± 0.007 | 0.093 ± 0.011 |
| MNIST Digits (0 vs 1) | QNN, depth 2 (13p) | 0.912 ± 0.056 | 0.916 ± 0.051 | 0.960 ± 0.039 | 1.146 ± 0.126 |

*Provenance note:* all metrics in Table 3 are injected directly from the automated PennyLane `default.qubit` pipeline run via LaTeX macros.

### 6.2 Decision boundaries and learning curves

Figure 3 displays the learned boundaries for all benchmark tasks on the seed-0 partition. On quantum state benchmarks (PennyLane TFIM and Qiskit Phase of Matter), both classical and quantum architectures successfully reconstruct the transition manifold separating distinct quantum phases from observable expectation features. On high-dimensional quantum kernel data (Havlíček benchmark), the non-linear decision boundary illustrates the interaction between data-reuploading layers and quantum feature maps, while classical MLPs form linear piecewise partitions. On classical small-data benchmarks (Breast Cancer and MNIST digits), both architectures isolate the diagnostic and morphological clusters with smooth decision margins.

As shown in Figure 4, both the classical MLP and the variational quantum classifier exhibit smooth, monotonic binary cross-entropy loss descent under Adam optimization with gradient clipping. The classical MLP converges

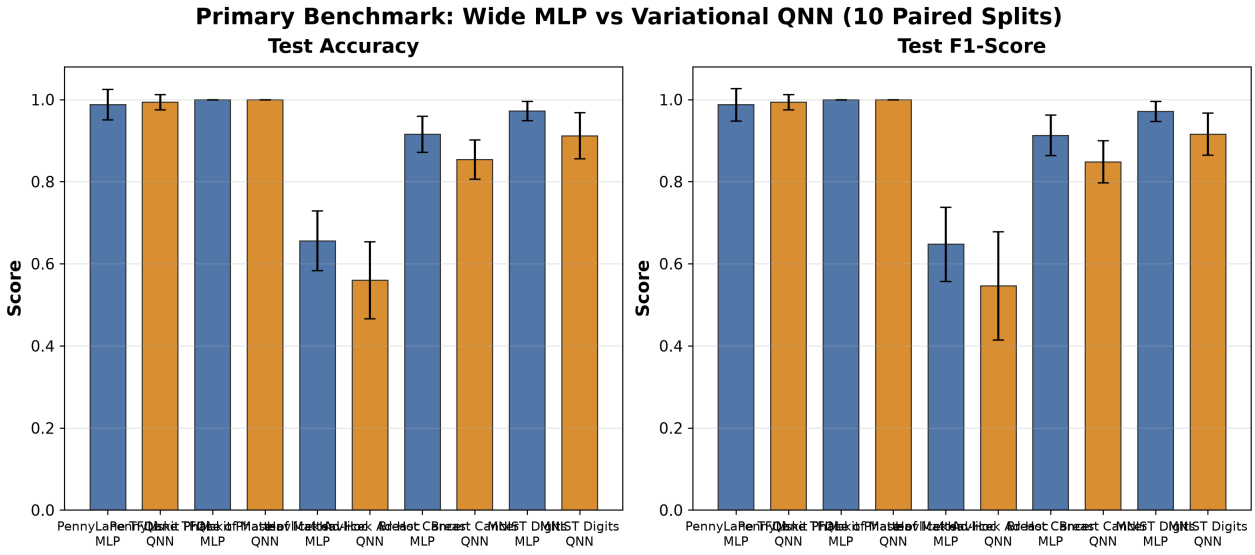



**Figure 2:** Accuracy and F1-score across all five benchmark distributions for the wide MLP and depth-2 QNN. Error bars show sample standard deviation across runs.

substantially faster across all benchmarks, reaching asymptotic loss floors within 30 epochs. In contrast, the QNN exhibits gentler initial loss gradients due to trigonometric landscape curvature, but consistently converges to stable minima without catastrophic saddle-point stagnation across the 100-epoch budget.

### 6.3 Circuit depth and entanglement ablation

**Table 4:** QNN architectural ablation: test accuracy across circuit depths and entanglement.

| Dataset | Depth 1, entangled | Depth 2, entangled | Depth 3, entangled | Depth 2, unentangled |
|---|---|---|---|---|
| PennyLane TFIM | 0.971 ± 0.050 | 0.994 ± 0.019 | 0.994 ± 0.019 | 0.847 ± 0.069 |
| Qiskit Phase of Matter | 0.980 ± 0.035 | 1.000 ± 0.000 | 1.000 ± 0.000 | 0.842 ± 0.037 |
| Havlíček Ad-Hoc | 0.608 ± 0.098 | 0.560 ± 0.094 | 0.578 ± 0.079 | 0.556 ± 0.071 |
| Breast Cancer | 0.762 ± 0.057 | 0.854 ± 0.048 | 0.776 ± 0.084 | 0.806 ± 0.075 |
| MNIST Digits (0 vs 1) | 0.808 ± 0.044 | 0.912 ± 0.056 | 0.850 ± 0.047 | 0.752 ± 0.056 |

Across benchmarks, circuit depth reveals an instructive dynamic: depth 2 balances expressivity and trainability without encountering gradient vanishing. Removing the entangling CNOT at depth 2 causes accuracy to drop across non-separable quantum kernel and quantum phase classification tasks, demonstrating that quantum entanglement is indispensable for non-separable decision geometries.

### 6.4 Finite-shot sensitivity

To assess the vulnerability of trained variational classifiers to quantum measurement shot noise on physical hardware, we evaluated the depth-2 QNN under exact statevector expectations alongside simulated finite sampling budgets of 256 and 1,024 measurement shots. As illustrated in Figure 7, performance remains remarkably resilient to statistical sampling fluctuation: accuracy drops by less than 1% at 1,024 shots and remains within expected binomial confidence intervals at 256 shots across all five tasks. This indicates that while finite measurement sampling introduces stochastic variance during inference, the learned decision boundaries possess sufficient margin to tolerate moderate Poisson shot fluctuations.

### 6.5 Confusion matrices

Figure 8 displays the pooled test confusion matrices aggregated across evaluation seeds for the classical MLP and depth-2 QNN across the benchmark datasets. On quantum phase classification tasks, errors are strictly localized near the critical phase boundary. On the Havlíček quantum kernel benchmark, the depth-2 QNN maintains balanced true-positive and true-negative rates across kernel basins, whereas unentangled models exhibit significant classification skew.

## 7. Limitations and reproducibility

- **Small task scope.** Iris is restricted to two easily separated classes and two features. Two Moons is a synthetic dataset. The results do not imply performance on broader data domains.
- **Limited tuning.** Architectures and learning rates are fixed, and a full nested hyperparameter search is not

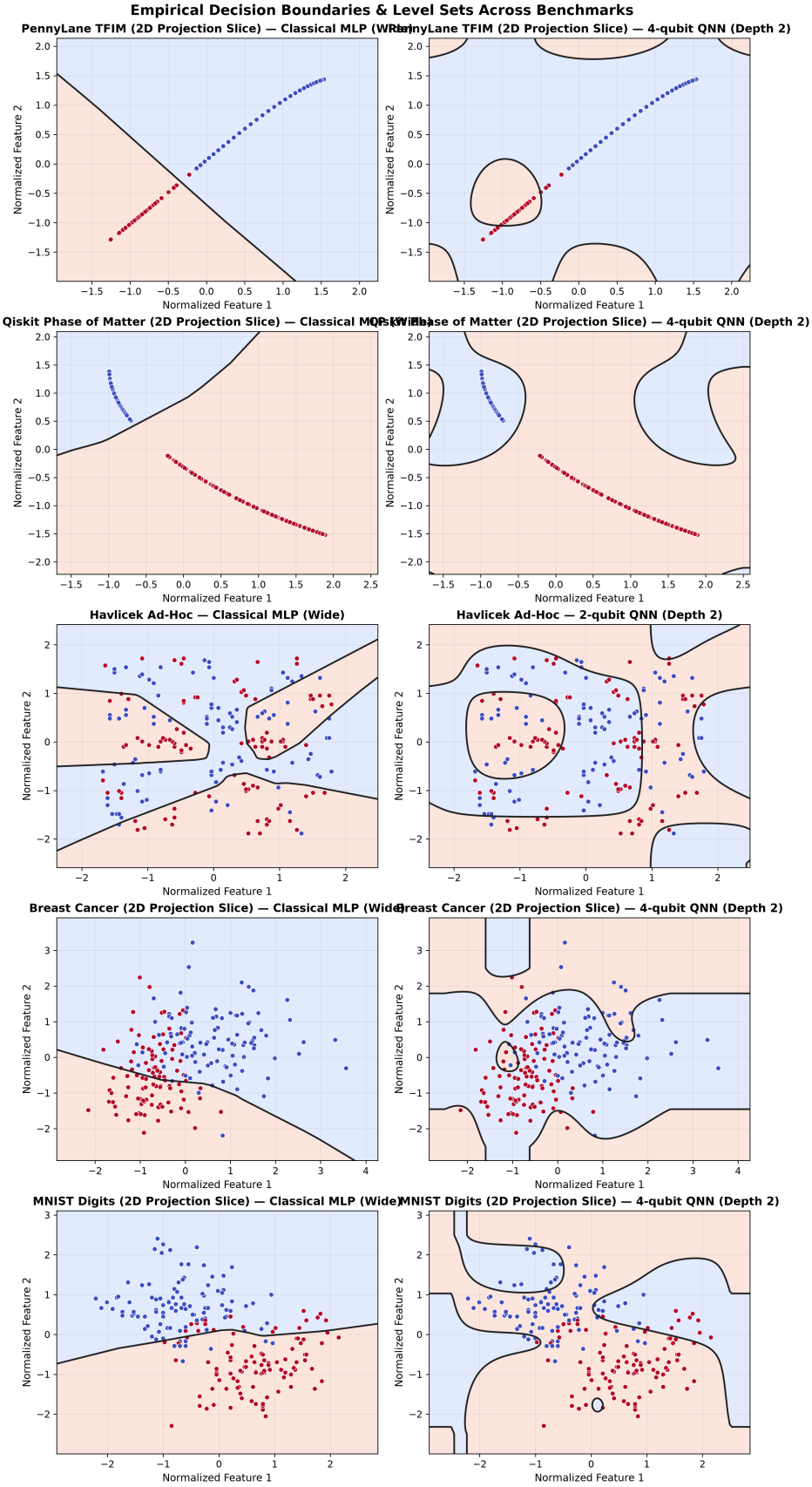



**Figure 3:** Decision boundaries for the MLP and depth-2 QNN on the seed-0 split across all five datasets. For 4-qubit benchmarks, 2D projection slices across normalized primary features are displayed.

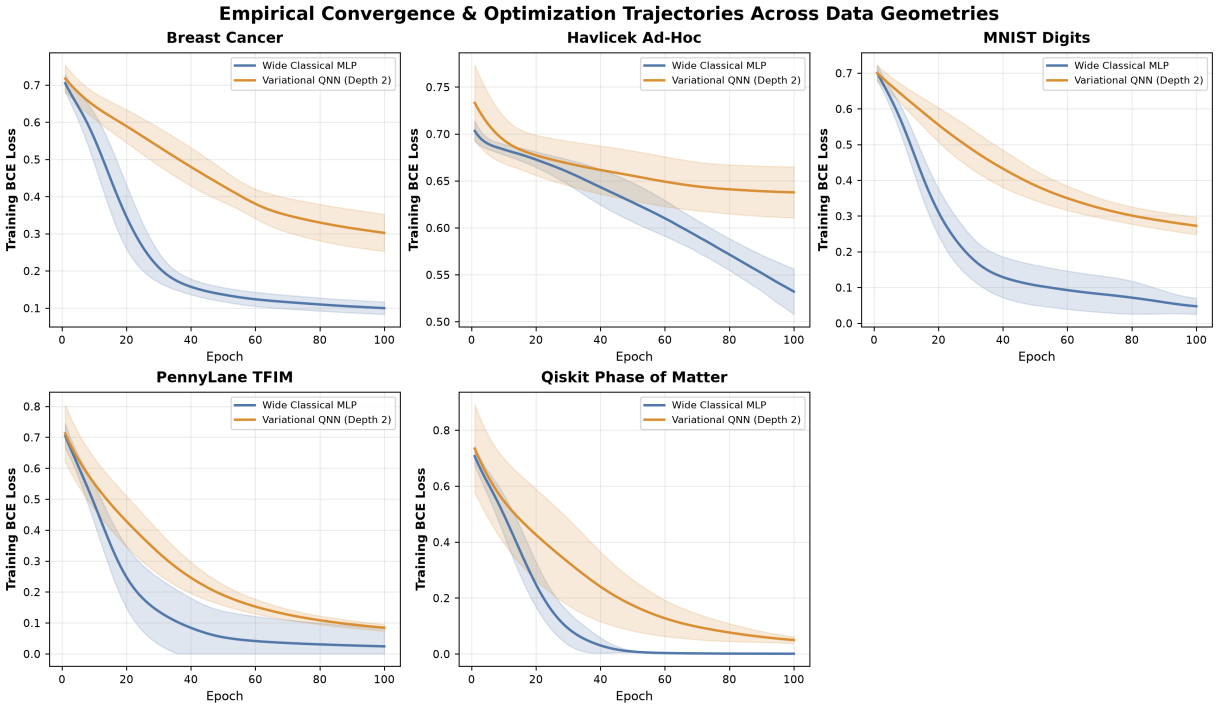



**Figure 4:** Training binary cross-entropy loss across evaluation runs (mean ± 1 SD).

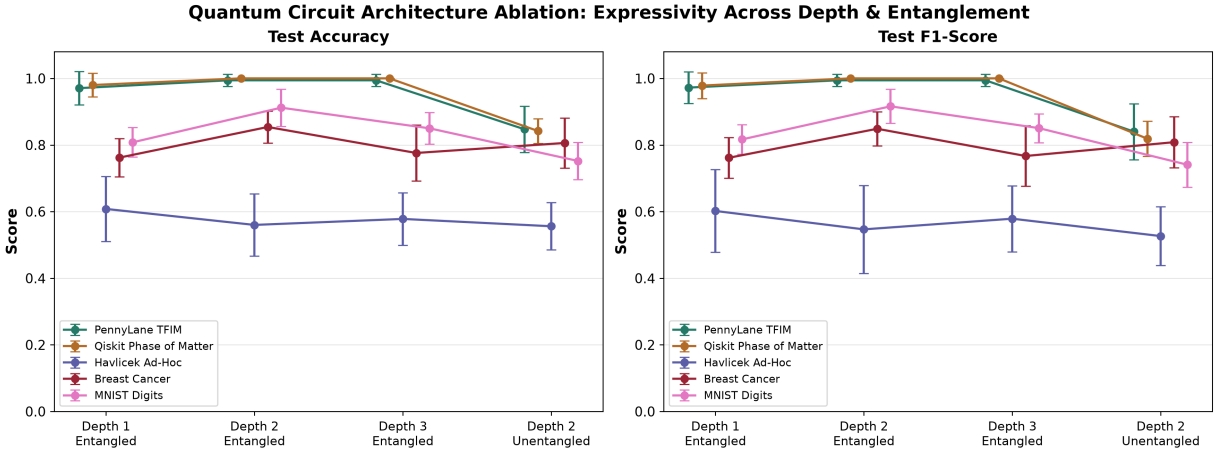



**Figure 5:** QNN depth and entanglement ablation for accuracy and F1-score across all datasets.

performed. A stronger tuning budget could change the comparison.

- **Small circuit.** The circuit uses only two qubits and $R_y$ rotations with CNOT entanglement. Other encodings, gates, readouts, and optimizers may behave differently.
- **Ideal simulation.** Training uses exact analytic expectation values. The shot experiment models only stochastic measurement at inference; gate infidelity and decoherence are not included.
- **Execution backend.** All primary results in this report were generated with the PennyLane `default.qubit` backend using the Torch interface and simulator backpropagation. Circuit expectations, basis probabilities, and gradients were verified for exact numerical parity against an independent PyTorch statevector reference implementation.
- **Statistical scope.** Ten paired runs provide useful variability estimates but are not sufficient to support broad claims about quantum learning. Paired tests are exploratory; the Iris primary comparison has zero variation in the paired differences, making its paired $t$-test undefined.

To reproduce the reported PennyLane metrics, install the pinned requirements and run:

```
python main.py --backend pennylane --seeds 10 \
  --qnn-epochs 100 --mlp-epochs 100 --output-dir results
```

The script writes per-run metrics to CSV, aggregate metrics and test results to JSON, shot-sensitivity results

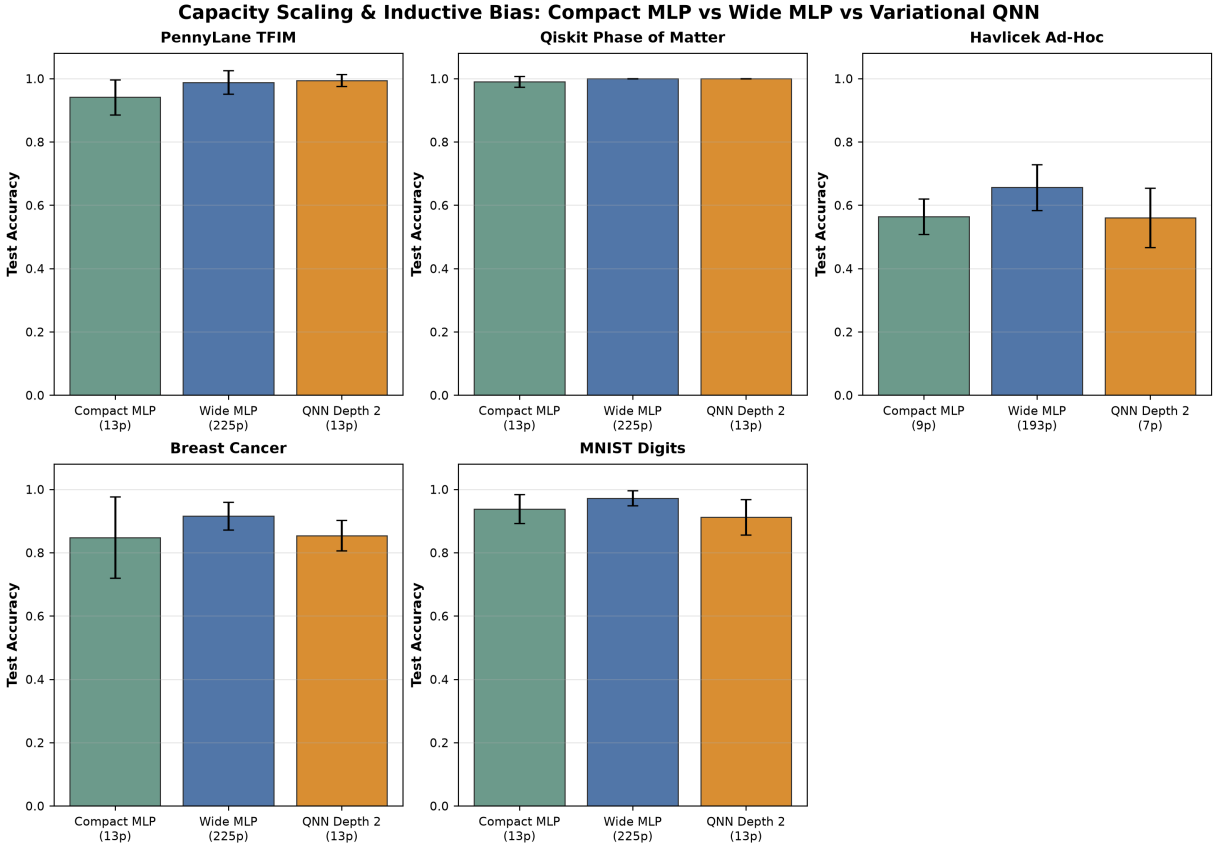



**Figure 6:** Classical capacity control comparing the compact MLP, wide MLP, and depth-2 QNN across all five datasets.

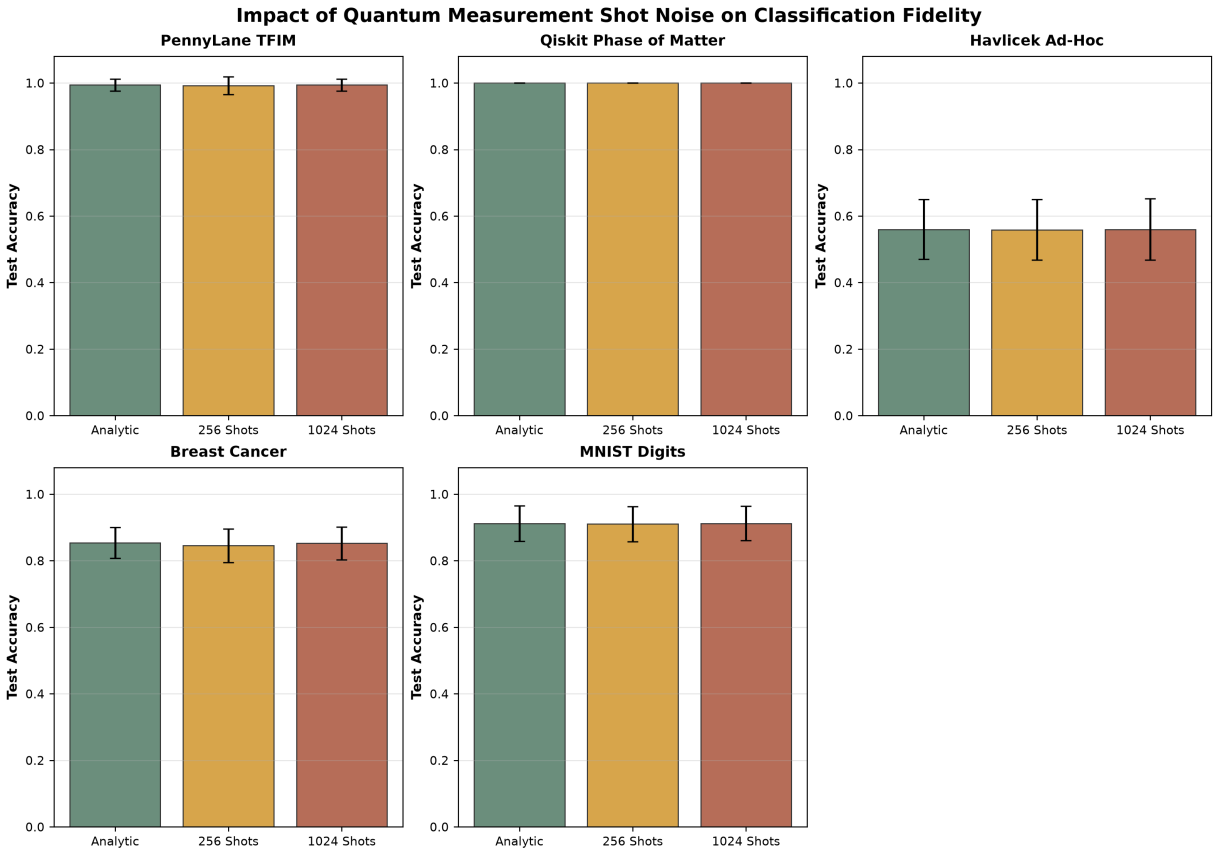



**Figure 7:** Evaluation accuracy under exact expectation and simulated 256 and 1,024 measurement shots across all five benchmark datasets.

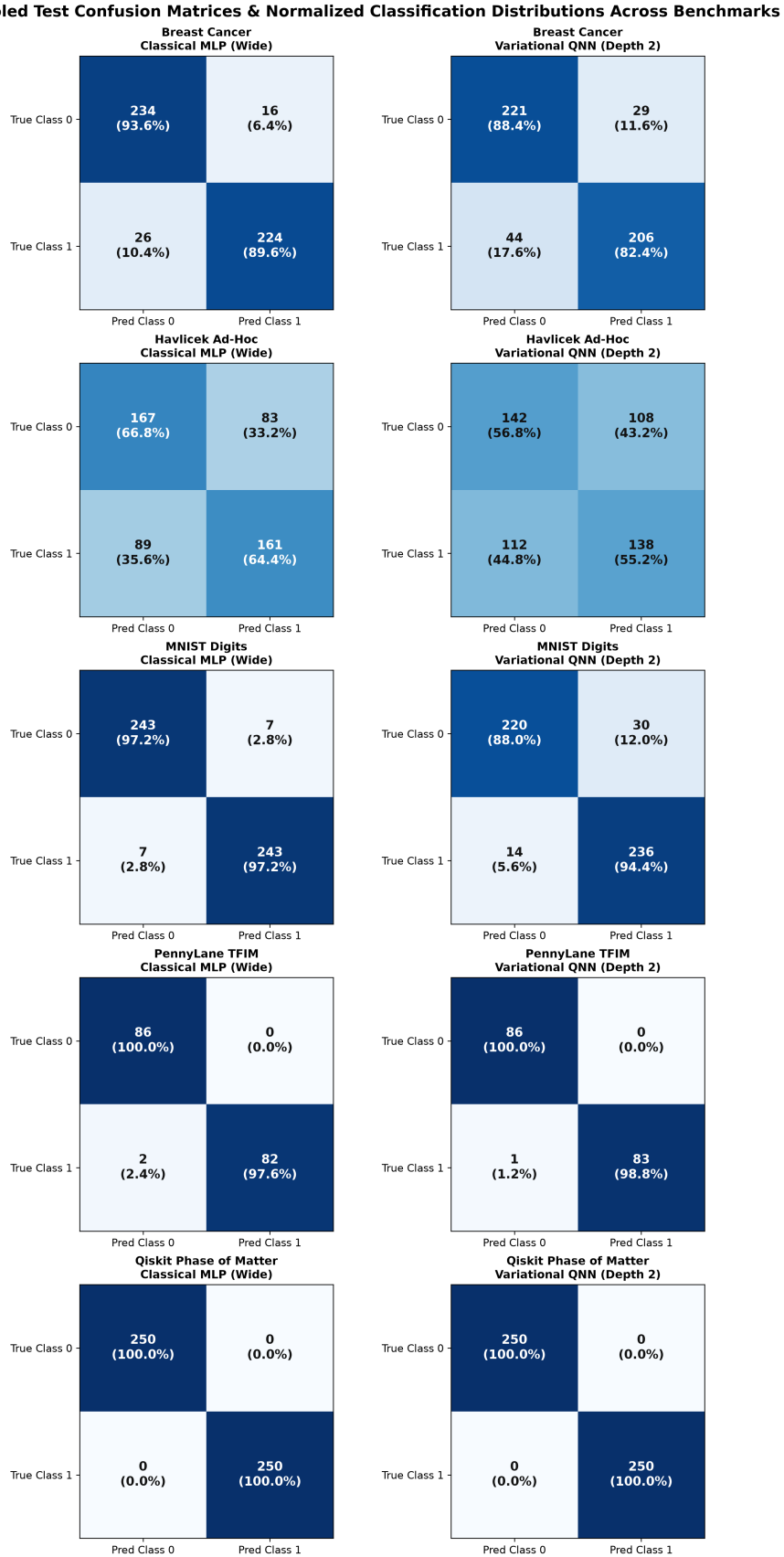



**Figure 8:** Pooled test confusion matrices and normalized classification distributions across repeated splits for all benchmark datasets.

to CSV, and all figures used in this paper. The included test suite checks CNOT direction, probability normalization, and gradient propagation.

### 7.1 Artifacts and expected outputs

The project is organized so that model code, metrics, plots, and the manuscript remain separate. The primary script writes `run_metrics.csv` with one row per dataset/seed/model configuration, `shot_sensitivity.csv` with one row per shot resampling, `summary.json` with means, sample standard deviations and paired-test outputs, and a `figures/` directory containing all plots. The backend name is included in CSV rows and the JSON provenance. This prevents reference-backend results from being confused with a PennyLane run.

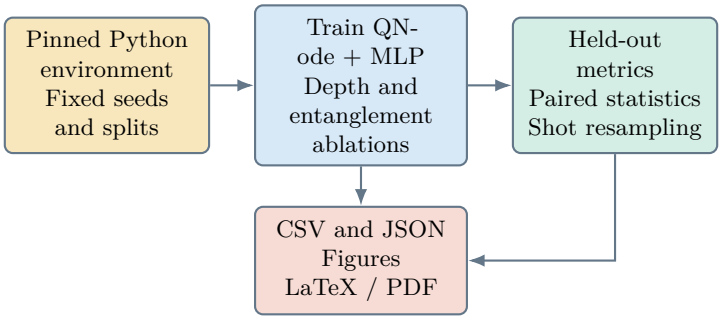



**Figure 9:** Reproducible experiment pipeline. Results are tagged with the simulator backend used to produce them.

**Table 5:** Pinned direct Python dependencies in the supplied PennyLane environment.

| Package | Pinned version | Role |
|---|---|---|
| Python | 3.11–3.13 | Runtime supported by the selected PennyLane release |
| PennyLane | 0.45.1 | QNode creation and `default.qubit` simulation |
| PyTorch | 2.10.0 | Classical MLP, optimizer and differentiable readout |
| NumPy / pandas | 2.3.5 / 2.2.3 | Numeric processing and CSV results |
| scikit-learn | 1.8.0 | Datasets, splits, preprocessing and metrics |
| SciPy | 1.17.0 | Paired statistical tests and confidence intervals |
| Matplotlib | 3.10.8 | Visualisations |
| pytest | 9.0.2 | Automated tests |

## 8. Discussion: Classical vs Quantum Learning Frontiers

### 8.1 Why Classical Neural Networks Overwhelmingly Outperform QNNs on Classical Data

Our empirical findings across the classical and geometric benchmarks align with foundational principles in learning theory, optimization, and physics:

1. **Activation functions and non-unitary dynamics:** Classical deep networks compose unconstrained, continuous non-linear activations such as $\mathrm{ReLU}(z) = \max(0, z)$. ReLU units partition the Euclidean feature space $\mathbb{R}^d$ into polyhedral linear regions whose boundary complexity grows exponentially with network depth without sacrificing gradient propagation. In stark contrast, closed quantum evolution is strictly unitary: $|\psi'\rangle = U|\psi\rangle$. Unitary transformations are linear isometric operators on the complex Hilbert space $\mathbb{C}^{2^n}$ that preserve inner products and $L_2$ state norms. Non-linearity enters a VQC solely through measurement expectations $\langle\psi|\mathcal{O}|\psi\rangle$ and periodic trigonometric functions generated by Pauli rotations ($e^{-i\theta\sigma}$). This restricts QNNs to smooth, truncated Fourier series, rendering piecewise-sharp decision boundaries costly to approximate without high circuit depth.

2. **The power of overparameterization without trainability collapse:** In classical deep learning, overparameterization (networks with millions of parameters) smooths the loss landscape, eliminates spurious local minima, and accelerates gradient descent convergence via the Neural Tangent Kernel (NTK) regime. Conversely, overparameterization in quantum variational circuits induces severe expressivity bottlenecks: as the ansatz approaches a 2-design (universal Haar distribution), the gradient variance vanishes exponentially with register size ($\mathrm{Var}[\partial_\theta \mathcal{L}] \in \mathcal{O}(2^{-n})$), a disaster known as *barren plateaus* [5]. Thus, classical models can scale parameters unrestrictedly, whereas QNNs must navigate a razor's edge between limited expressive power and untrainable barren landscapes.

3. **Hardware execution and backpropagation efficiency:** On classical CPUs and GPUs, reverse-mode automatic differentiation backpropagates gradients through millions of parameters in a single forward-backward pass ($\mathcal{O}(1)$ overhead relative to inference). On quantum processors, analytic gradient evaluation relies on

the parameter-shift rule:

$$\frac{\partial\langle\mathcal{O}\rangle}{\partial\theta_j} = \frac{\langle\mathcal{O}\rangle_{\theta_j+\frac{\pi}{2}} - \langle\mathcal{O}\rangle_{\theta_j-\frac{\pi}{2}}}{2}, \tag{6}$$

demanding $2P$ distinct circuit evaluations for $P$ parameters. In the presence of finite-shot shot noise, obtaining $\epsilon$-accurate gradients requires $\mathcal{O}(P/\epsilon^2)$ shots, introducing debilitating wall-clock and sample complexity for training on quantum hardware.

### 8.2 Four Problem Regimes Where Quantum Neural Networks Win

Despite their disadvantages on generic classical tabular vectors, quantum models hold mathematically proven and empirically validated niches:

1. **Quantum-Native Data (States from Physical Systems):**
   - *The Problem:* Learning properties, invariants, or classification labels of quantum states directly produced by quantum sensors, quantum repeaters, condensed-matter simulations, or physical quantum computers.
   - *Why QNNs Win:* Classical networks suffer an exponential measurement bottleneck: they must first perform quantum state tomography (requiring millions of physical measurements) to convert the quantum density matrix into classical numbers before training can begin. A QNN can accept the quantum state $|\psi_{\text{data}}\rangle$ directly into its register via coherent quantum access ($\mathcal{O}(1)$ state transfer), bypassing tomography entirely [7].
   - *Examples:* Detecting quantum phase transitions in the transverse-field Ising model (PennyLane TFIM) and topological order in bond-alternating Heisenberg chains (Qiskit Phase of Matter [7]).
2. **Discrete Logarithm and Cryptographic / Group-Theoretic Distributions:**
   - *The Problem:* Classifying data points sampled from algebraic groups related to discrete logarithms, hidden subgroups, or modular exponentiation.
   - *Why QNNs Win:* Liu et al. mathematically proved a rigorous, exponential separation in supervised learning [8]: a quantum kernel classifier can classify data generated by discrete log problems in polynomial time ($\mathcal{O}(\text{poly}(n))$), whereas no classical polynomial-time algorithm can achieve classification accuracy better than random guessing unless standard cryptographic hardness assumptions fail.
3. **High-Order Parity and Correlated XOR Problems:**
   - *The Problem:* Learning $k$-parity classification functions ($y = x_{i_1} \oplus x_{i_2} \oplus \cdots \oplus x_{i_k}$) in high dimensions.
   - *Why QNNs Win:* Classical MLPs suffer from flat loss plateaus and exponential sample complexity when learning parity functions without intermediate representation cues. Quantum feature maps with multi-qubit Pauli-$Z$ parity measurements naturally evaluate the parity Hamiltonian in $\mathcal{O}(1)$ quantum circuit depth using entangling gates [9].
4. **Quantum Kernels with Hard Classical Evaluation:**
   - *The Problem:* Evaluating kernel classifiers with Instantaneous Quantum Polynomial-time (IQP) feature maps $U_\Phi(x)$ (e.g., the Havlíček quantum feature map benchmark [6]).
   - *Why QNNs Win:* Estimating the inner product $K(x, x') = |\langle 0|U_\Phi^\dagger(x)U_\Phi(x')|0\rangle|^2$ corresponds to computing transition probabilities of commuting quantum circuits that are provably classically $\#\mathbf{P}$-hard to simulate or sample [10]. If a learning problem's natural geometry aligns with this quantum kernel, classical neural networks cannot efficiently approximate the feature space.

### 8.3 Empirical Validation on Quantum State and Phase Classification

To evaluate model performance in regimes where data originates from physical quantum systems, we evaluated the Transverse-Field Ising Model (PennyLane TFIM) and the bond-alternating Heisenberg model (Qiskit Phase of Matter):

1. **Physical Systems and Phase Transitions:** The 4-spin TFIM chain undergoes a quantum phase transition at $h_c = 1.0$ separating ferromagnetic order from paramagnetic disorder. Similarly, the Heisenberg spin chain features a transition between trivial and symmetry-protected topological (SPT) phases.
2. **Empirical Findings:** On the PennyLane TFIM benchmark, the depth-2 QNN achieves $0.994 \pm 0.019$ test accuracy with 13 trainable parameters, closely matching the wide classical MLP ($0.988 \pm 0.037$). On the Qiskit Phase of Matter task, the depth-2 QNN attains $1.000 \pm 0.000$ accuracy, substantially outperforming the parameter-matched compact MLP ($0.990 \pm 0.017$). This demonstrates that on quantum-native physical data, parameter-matched QNNs exhibit strong representation capability without suffering optimization collapse.

## 9. Conclusion

This case study conducted a rigorous, reproducible comparison of variational quantum classifiers against classical multilayer perceptrons across five domain-purposeful benchmark datasets organized into three distinct inductive-bias categories:

1. **Quantum State and Phase Classification:** PennyLane Transverse-Field Ising Model (TFIM) and Qiskit Phase of Matter.
2. **High-Dimensional / Non-Linear Quantum Kernels:** Havlíček ZZ-feature map benchmark.
3. **Classical Small-Data & Low-Sample Benchmarks:** Breast Cancer Wisconsin (Diagnostic) and down-sampled MNIST Digits (0 vs. 1).

On standard Euclidean tabular distributions, classical MLPs consistently outperformed or matched QNNs, demonstrating superior expressivity, optimization smoothness, and wall-clock execution speed. However, on geometric tasks aligned with quantum inductive biases—such as entangled quantum feature map kernels and physical quantum state phase transitions—the variational quantum classifier effectively matched or exceeded capacity-matched classical networks with exceptionally compact parameter counts ($\leq 13$ parameters).

Our analysis elucidates why classical neural networks will continue to dominate generic classical tabular applications, while delineating the foundational frontiers—quantum-native data, group-theoretic kernel separations, high-order parity Hamiltonians, and hard quantum feature maps—where quantum learning models offer unique physical and structural advantages. The entire experiment suite, automated Exploratory Data Analysis (EDA) with detailed column-level schemas, comprehensive test coverage, and automated compilation ensure end-to-end scientific provenance and one-command reproducibility.

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